Analytics in HR for Recruitment, Training, Performance Evaluation and Attrition Prediction

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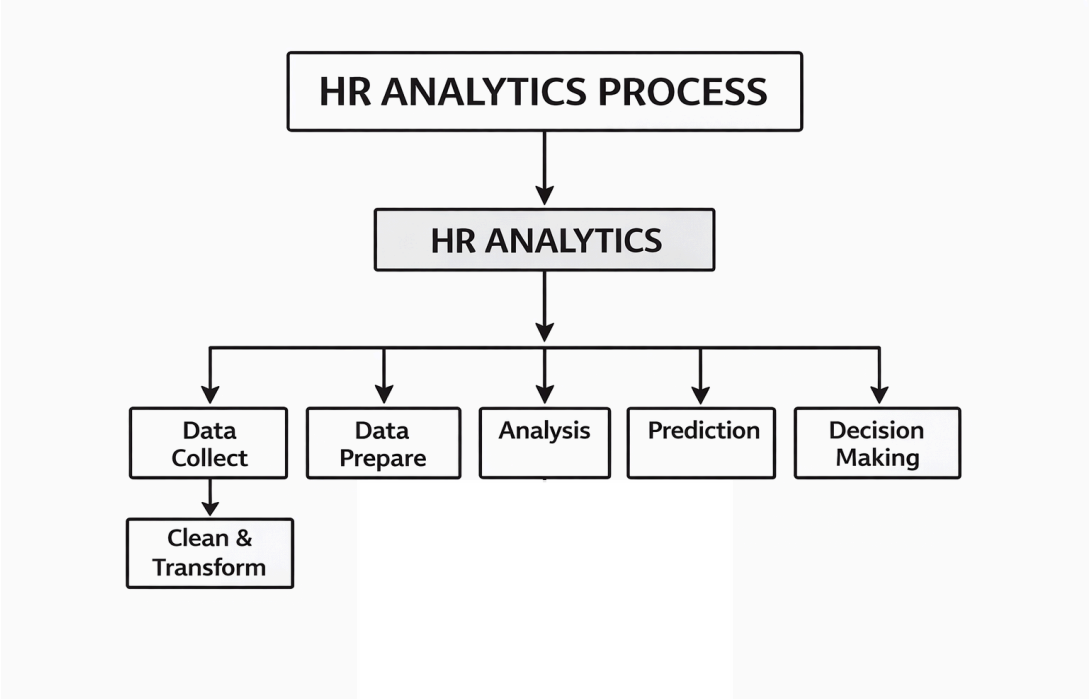
Analytics in HR

Analytics in HR means using data, statistical methods, and models to make better decisions about employees. It helps organizations to understand employee behavior, improve performance, and predict future outcomes.

According to the notes, analytics follows a process:

- Data collection
- Data preparation
- Analysis
- Prediction
- Decision making



This process is applied in HR functions as follows:

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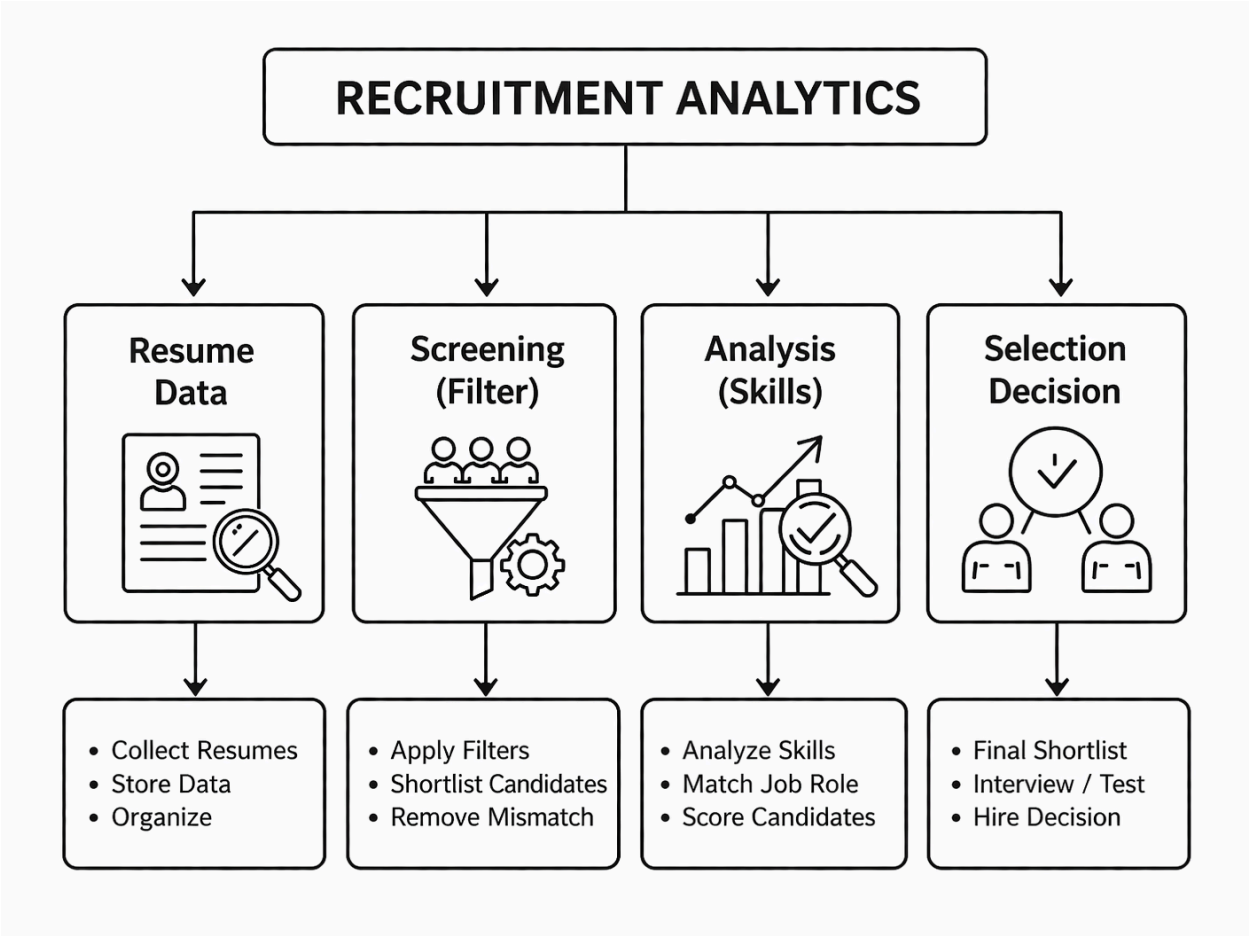
1. Analytics in Recruitment

Using data to identify and select the best candidates.



Application:

- ❖ Collect candidate data (resume, test scores, interview results)
- ❖ Clean and prepare data before analysis
- ❖ Use descriptive analytics → understand past hiring performance
- ❖ Use predictive analytics → predict candidate success
- ❖ Use data filters to shortlist candidates

Example:

- ❖ If data shows candidates with certain skills perform better, company prefers those skills

Benefits:

- ❖ Faster hiring
- ❖ Better candidate selection
- ❖ Reduces bias
- ❖ Improves employee quality

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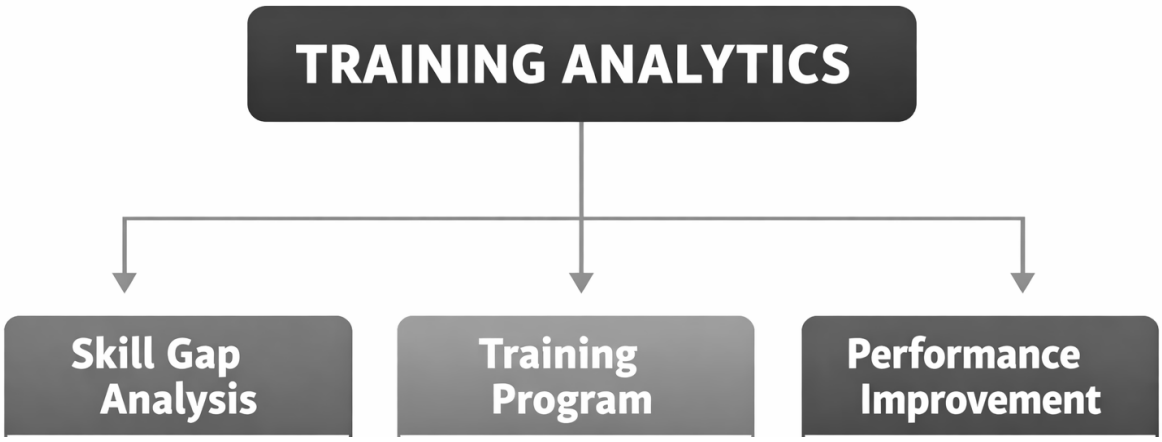
2. Analytics in Training and Development

Utilizing data-driven insights to refine employee competencies and enhance the efficacy of instructional programs.



Application:

- ❖ Evaluate performance metrics → before and after instructional sessions
- ❖ Pinpoint skill deficiencies utilizing employee data sets
- ❖ Use diagnostic analytics → determine root causes of low productivity
- ❖ Deliver targeted training initiatives
- ❖ Quantify and measure training outcomes

Example:

- If assessments identify a weakness in communication, specific training is allocated to that domain

Benefits:

- ❖ Enhances workforce skillsets
- ❖ Optimizes instructional expenditures
- ❖ Boosts overall organizational productivity

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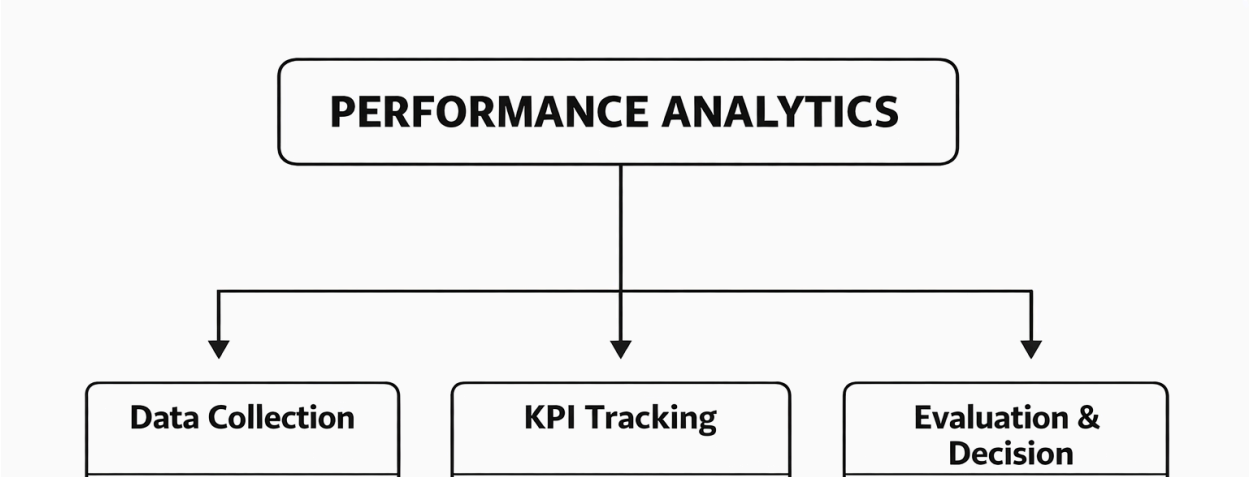
3. Analytics in Performance Evaluation

Implementing data-driven methodologies to quantify and monitor employee performance levels.



Application:

- ❖ Aggregate data points → including attendance, targets, and feedback
- ❖ Utilize KPIs (Key Performance Indicators)
- ❖ Apply descriptive analytics → evaluate historical performance outcomes
- ❖ Apply diagnostic analytics → identify causes of performance fluctuations
- ❖ Perform benchmarking → compare performance across the workforce

Example:

- Sales personnel performance is quantified through the analysis of sales data

Benefits:

- ❖ Ensures objective and fair evaluations

- ❖ Supports data-backed promotion and compensation decisions
- ❖ Increases employee engagement and motivation

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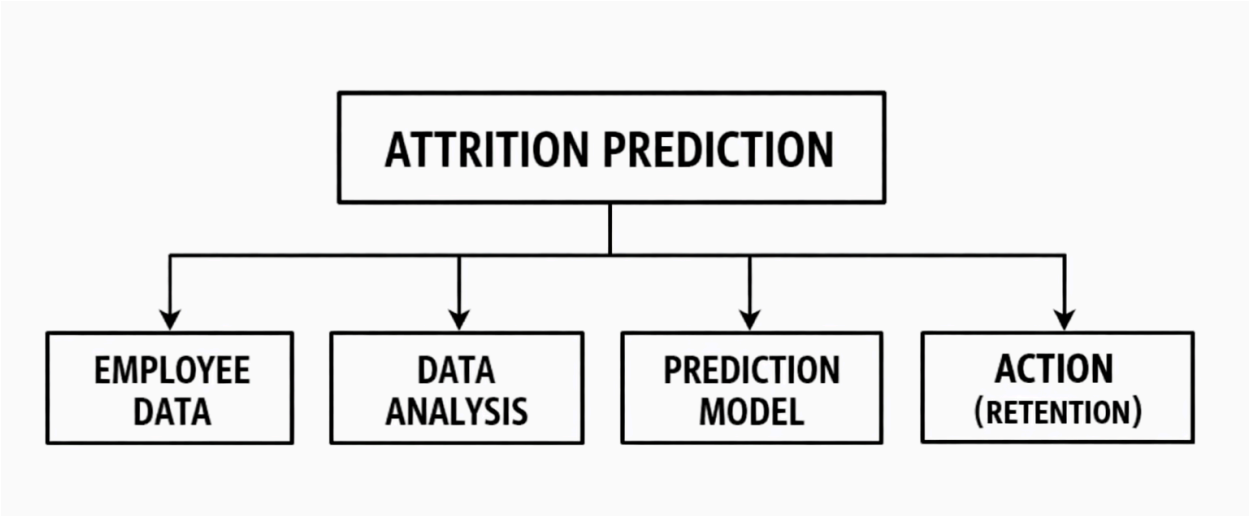
4. Analytics in Attrition Prediction

Forecasting personnel at risk of departure to optimize retention strategies.



Application:

- ❖ Analyze historical employee data sets
- ❖ Evaluate influencing factors such as:
  - ➢ Compensation structures
  - ➢ Workplace environment
  - ➢ Job satisfaction levels
  - ➢ Career progression and promotion history
- ❖ Utilize predictive models → forecast potential attrition
- ❖ Implement proactive preventive measures

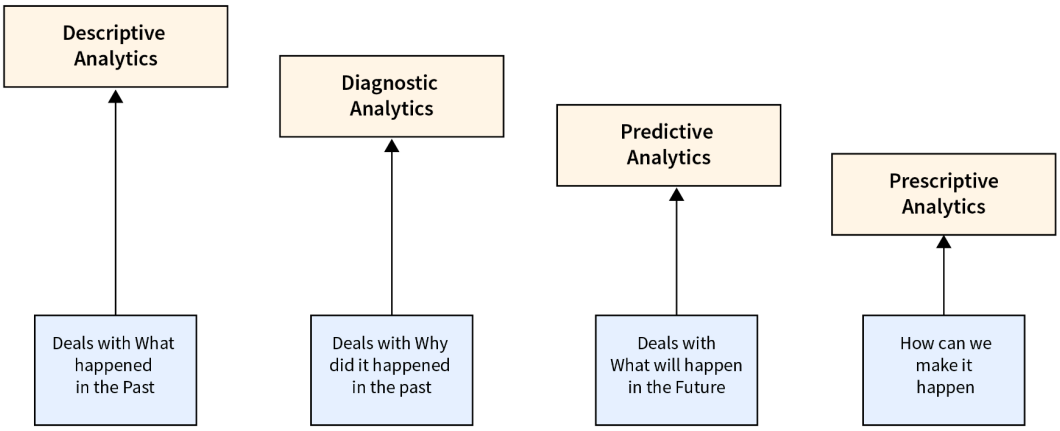
Example:

- Employees with low satisfaction metrics and stagnant promotion history represent a high turnover risk

Benefits:

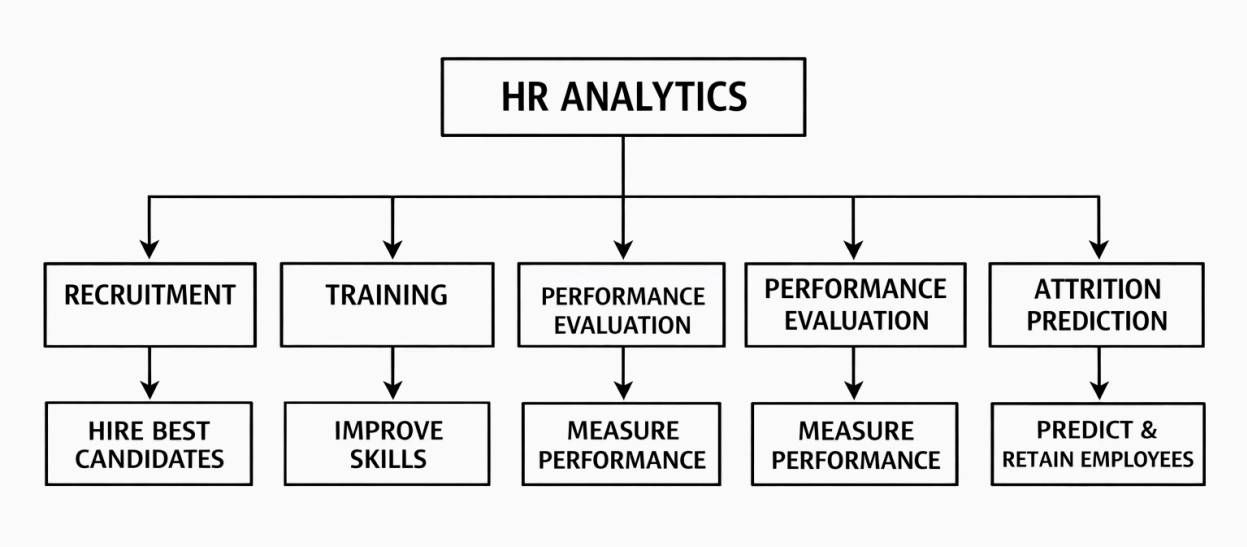
- Reduces employee turnover rates
- Minimizes recruitment and onboarding costs
- Strengthens overall employee retention

Types of Analytics Used in HR



Key Points

- HR Analytics uses data for decision making
- Helps in hiring, training, evaluation, and retention
- Uses descriptive, diagnostic, predictive, and prescriptive analytics
- Improves efficiency and reduces cost
- Supports better planning and future prediction



SUPPLY CHAIN NETWORK PLANNING

Conceptual Overview

Supply Chain Network Planning involves the strategic **architectural design and management of a supply chain framework**. It encompasses pivotal decisions regarding:

- Strategic placement of manufacturing plants, warehouses, and distribution centers
- Orchestrating the flow of commodities: supplier → manufacturer → warehouse → consumer
- Mitigating operational expenditures while enhancing service efficacy

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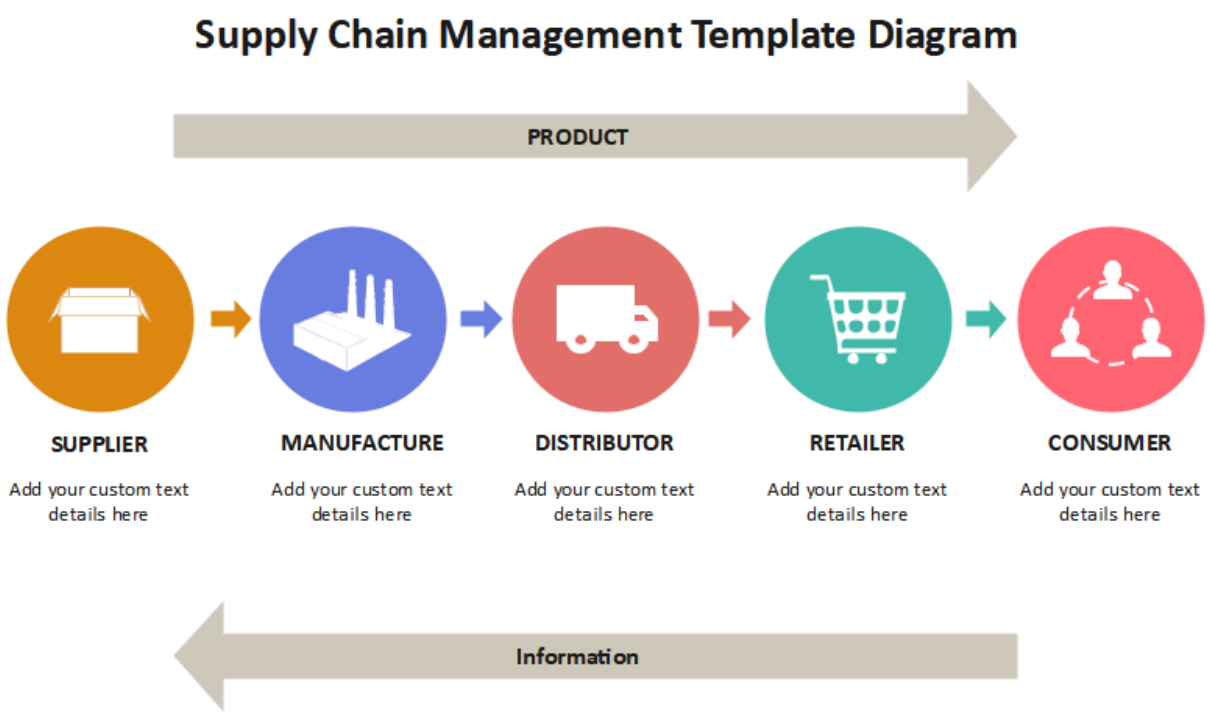
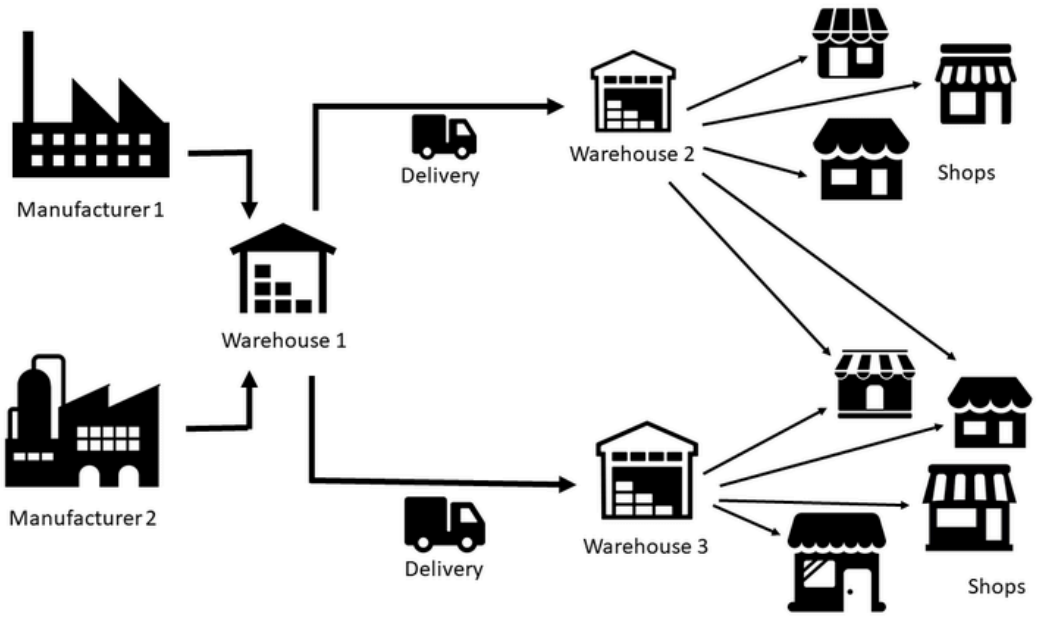
Architecture of a Supply Chain Network



Operational Flow Analysis:

- Supplier → provision of raw materials
- Manufacturer → production of finished goods
- Warehouse → commodity storage and inventory management
- Distributor/Retailer → facilitates sales to the consumer
- Customer → terminal end-user



Strategic Objectives of Network Planning

- Minimize total expenditures (transportation, storage, and production)
- Accelerate delivery velocity and responsiveness
- Effectively fulfill consumer demand requirements
- Optimize resource allocation and utilization
- Diminish systemic risks and operational latencies

Key Strategic Decisions in Network Planning

- **Locational Strategy**
  - Determination of optimal sites for warehouses and manufacturing facilities
- **Capacity Analytics**
  - Quantifying required production and storage volumes
- **Logistical Frameworks**
  - Identifying optimal routing and modes (road, rail, air)
- **Inventory Positioning**
  - Strategic placement and quantification of stock

ROLE OF ANALYTICS IN SUPPLY CHAIN MANAGEMENT

Analytics facilitates **data-driven decision-making** to replace heuristic-based estimation.

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1. ANALYTICS IN DEMAND PLANNING

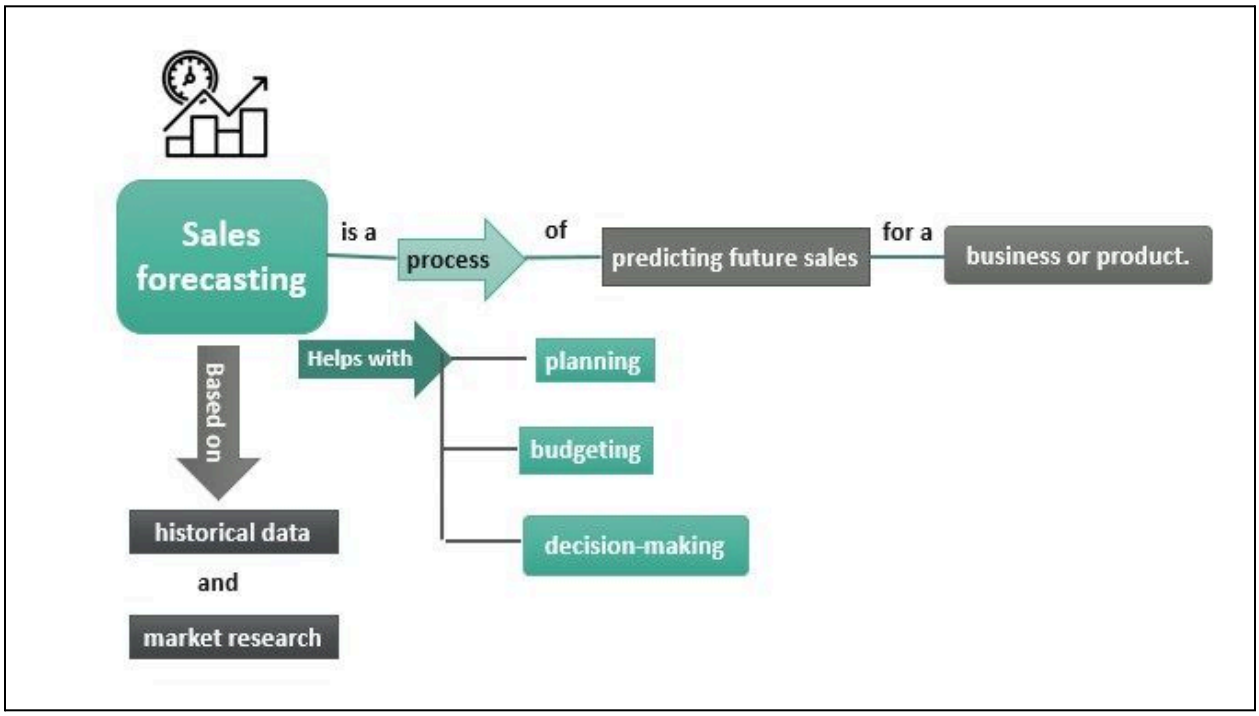
Application

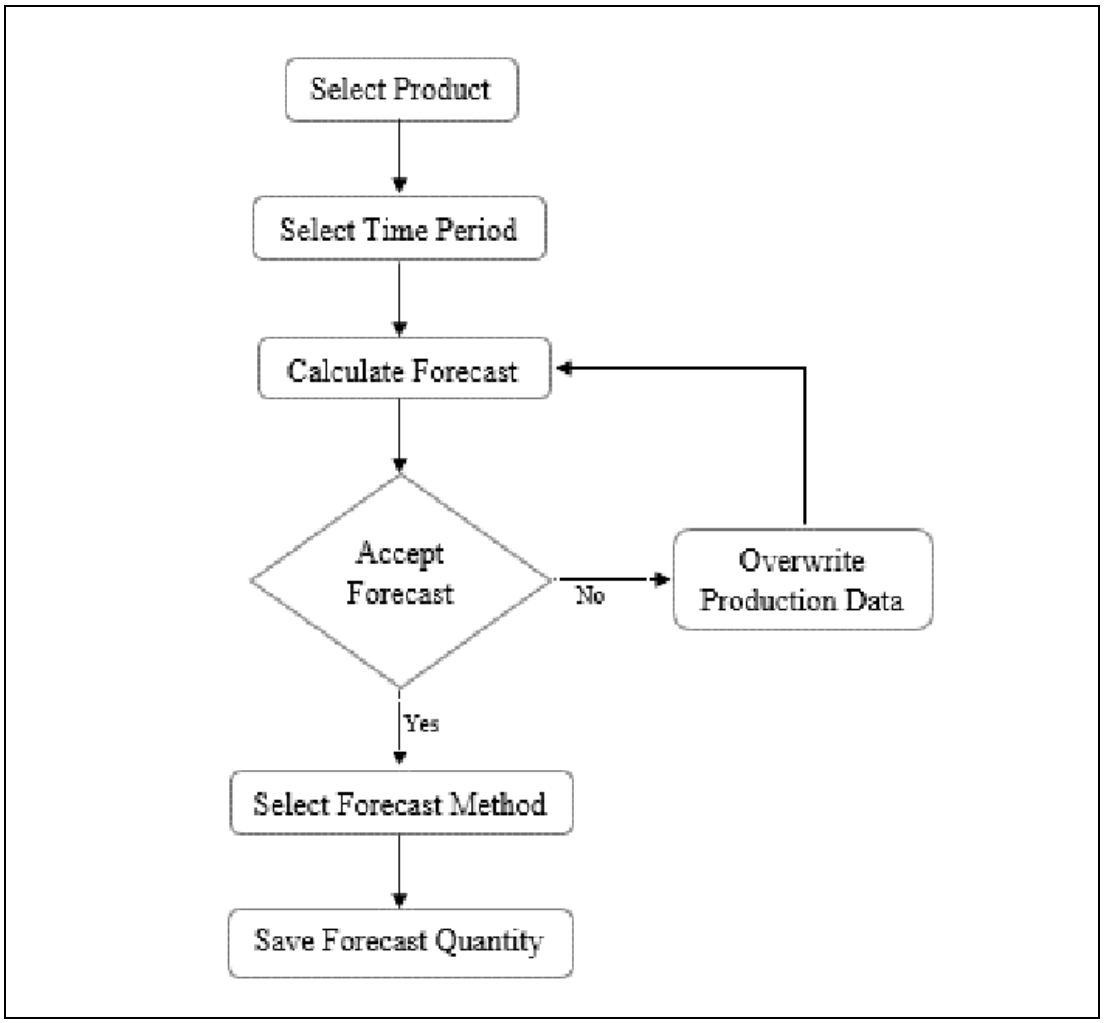
Demand planning forecasts **future customer requirements** utilizing historical data sets.

Techniques Utilized

- Time series analytical modeling
- Machine learning predictive models
- Historical trend evaluation
- Seasonal forecasting frameworks

Demand Planning Workflow





Analytical Contributions

- Generate accurate future demand projections
- Pinpoint seasonal fluctuation patterns
- Mitigate inventory imbalances, including overstock and stockouts
- Refine and optimize production scheduling

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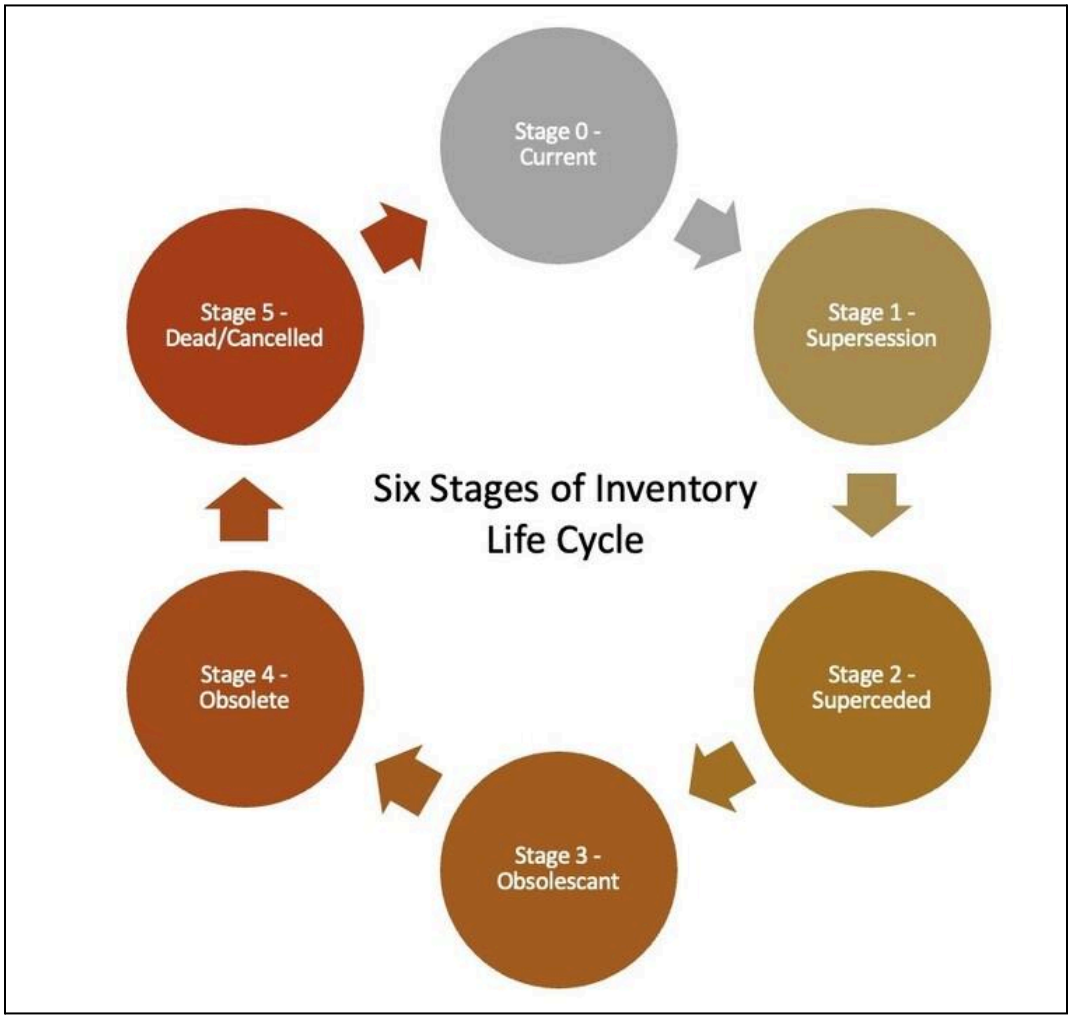
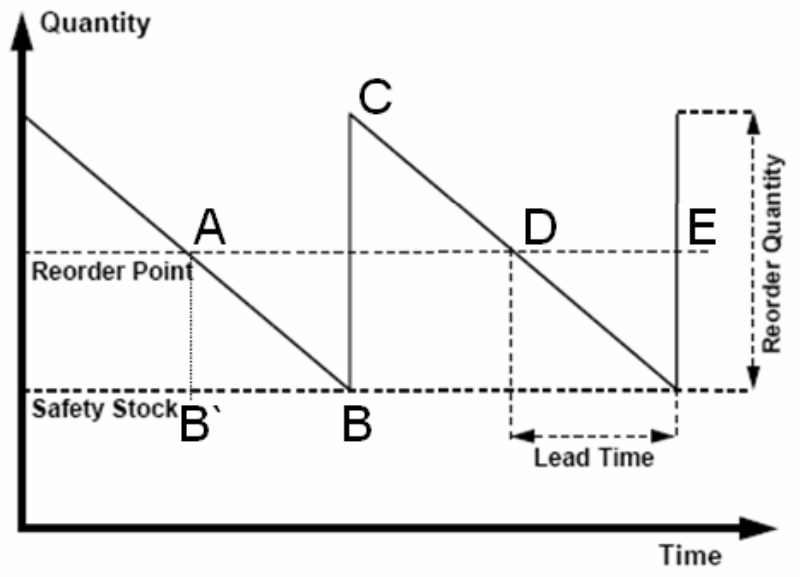
2. ANALYTICS IN INVENTORY MANAGEMENT

Conceptual Overview

Inventory analytics assists in determining **optimal stock levels and precise reorder schedules**.

Inventory Operational Flow





Analytical Contributions

- Quantify optimal inventory volumes (EOQ)
- Mitigate holding and carrying costs
- Avert stock shortages and supply gaps
- Enhance warehouse operational efficiency

Methodologies

- Economic Order Quantity (EOQ) modeling
- ABC Classification Analysis
- Safety stock quantification
- Reorder point analytical evaluation

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3. ANALYTICS IN LOGISTICS

Conceptual Overview

Logistics entails the strategic **coordination of transport and commodity distribution**.

Logistical Workflow

Analytical Contributions

- Execute shortest-path route optimization
- Mitigate fuel consumption and transportation expenditures
- Enhance delivery velocity and temporal accuracy
- Implement real-time shipment monitoring and tracking

Methodologies

- Vehicle Routing Problem (VRP) analytical modeling
- GPS integration with real-time analytical monitoring
- Simulation modeling architectures
- Advanced optimization algorithmic frameworks